

Polar Decomposition

Any $\underline{\underline{F}}$ with $\det \underline{\underline{F}} = \frac{dV_x}{dV_x} > 0$ has

right & left polar decomposition $\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}} = \underline{\underline{V}} \underline{\underline{R}}$

where

$\underline{\underline{R}}$ = rotation

$$\left. \begin{aligned} \underline{\underline{U}} &= \sqrt{\underline{\underline{F}}^T \underline{\underline{F}}} \\ \underline{\underline{V}} &= \sqrt{\underline{\underline{F}} \underline{\underline{F}}^T} \end{aligned} \right\} \begin{aligned} &\text{symmetric positive definite} \\ &\underline{\underline{a}} \cdot \underline{\underline{U}} \underline{\underline{a}} > 0 \\ &\underline{\underline{a}} \cdot \underline{\underline{V}} \underline{\underline{a}} > 0 \end{aligned} \quad \text{for all } \underline{\underline{a}} \in \mathcal{V}$$

$\Rightarrow$ eigenvalues of $\underline{\underline{U}}$ and $\underline{\underline{V}}$ are positive: $\lambda_i > 0$

$$\underline{\underline{F}} \underline{\underline{U}}^{-1} = \underline{\underline{R}} \underline{\underline{U}} \underline{\underline{U}}^{-1} \Rightarrow \underline{\underline{R}} = \underline{\underline{F}} \underline{\underline{U}}^{-1}$$

note $\uparrow$ to self:
yes there will
be λ 's!

For symmetric tensors: $\underline{\underline{A}} = \underline{\underline{A}}^T \Rightarrow (\underline{\underline{A}}^T)^{-1} = (\underline{\underline{A}}^{-1})^T$

$$\underline{\underline{A}}^{-1} \underline{\underline{A}} = \underline{\underline{I}}$$

$$\textcircled{1} (\underline{\underline{A}}^T)^{-1} \underline{\underline{A}} = \underline{\underline{I}}$$

$$\underline{\underline{A}} \underline{\underline{A}}^{-1} = \underline{\underline{I}}$$

$$(\underline{\underline{A}} \underline{\underline{A}}^{-1})^T = \underline{\underline{I}}^T$$

$$(\underline{\underline{A}}^{-1})^T \underline{\underline{A}}^T = \underline{\underline{I}}$$

$$\textcircled{2} (\underline{\underline{A}}^{-1})^T \underline{\underline{A}} = \underline{\underline{I}}$$

$$1+2: (\underline{\underline{A}}^T)^{-1} \underline{\underline{A}} = (\underline{\underline{A}}^{-1})^T \underline{\underline{A}} \Rightarrow (\underline{\underline{A}}^T)^{-1} = (\underline{\underline{A}}^{-1})^T$$

Show $\underline{\underline{R}}$ is rotation:

1) $\underline{\underline{R}}^T \underline{\underline{R}} = \underline{\underline{R}} \underline{\underline{R}}^T = \underline{\underline{I}}$ (orthonormal)

2) $\det(\underline{\underline{R}}) = 1$ (rotation)

1) Orthonormal

$$\begin{aligned} \underline{\underline{R}}^T \underline{\underline{R}} &= (\underline{\underline{F}} \underline{\underline{U}}^{-1})^T (\underline{\underline{F}} \underline{\underline{U}}^{-1}) = (\underline{\underline{U}}^{-1})^T \underline{\underline{F}}^T \underline{\underline{F}} \underline{\underline{U}}^{-1} = (\underline{\underline{U}}^T)^{-1} \underbrace{\underline{\underline{F}}^T \underline{\underline{F}}}_{\underline{\underline{U}}^2} \underline{\underline{U}}^{-1} \\ &= \underbrace{\underline{\underline{U}}^{-1} \underline{\underline{U}}^2}_{\underline{\underline{I}}} \underbrace{\underline{\underline{U}}^{-1}}_{\underline{\underline{I}}} = \underline{\underline{I}} \quad \checkmark \quad \begin{array}{c} \uparrow \\ \text{sym.} \end{array} \end{aligned}$$

2) Rotation

$$\det(\underline{\underline{R}}) = \det(\underline{\underline{F}} \underline{\underline{U}}^{-1}) = \frac{\det(\underline{\underline{F}})}{\det(\underline{\underline{U}})} > 0$$

$$\det(\underline{\underline{F}}) = \frac{dV_x}{dV_X} > 0 \text{ for admissible deformation}$$

$$\det(\underline{\underline{U}}) = \lambda_1 \lambda_2 \lambda_3 > 0 \text{ if } \underline{\underline{U}} \text{ is s.p.d.}$$

Show $\underline{U}$ is symmetric positive definite

$$|\underline{a}|^2 = \underline{a} \cdot \underline{a} > 0$$

$$\underline{a} = \underline{F} \underline{v}$$

$$(\underline{F} \underline{v}) \cdot (\underline{F} \underline{v}) > 0$$

$$= \underline{v} \cdot \underline{F}^T \underline{F} \underline{v} = \underline{v} \cdot \underline{U}^2 \underline{v} = \underline{v} \cdot \underline{C} \underline{v} > 0$$

$$\Rightarrow \underline{F}^T \underline{F} = \underline{U}^2 = \underline{C} \quad \text{is s.p.d.}$$

eigenvalues of $\underline{C}$: $\mu_i > 0$

↑ yes that's what they will be called!

Is $\underline{U} = \sqrt{\underline{C}}$ also s.p.d.?

Tensor square root

If $\underline{C}$ is a s.p.d. tensor with eigenpair $(\mu_i, \underline{u}_i)$

$$\underline{U} = \sqrt{\underline{C}} = \sum_{i=1}^3 \sqrt{\mu_i} \underline{u}_i \otimes \underline{u}_i$$

eigenpair of $\underline{U}$ is $(\lambda, \underline{u})$ where $\lambda_i = \sqrt{\mu_i} > 0$

$$\Rightarrow \underline{U} = \sqrt{\underline{C}} = \sqrt{\underline{F}^T \underline{F}} \quad \text{is s.p.d.}$$

Similarly $\underline{V} = \sqrt{\underline{F} \underline{F}^T}$ is s.p.d.

Analysis of local deformation

locally any $\varphi(\underline{x})$ can be approximated as

homogeneous $\underline{x} = \varphi(\underline{x}) = \underline{c} + \underline{\underline{F}} \underline{x} \quad \underline{\underline{F}} = \nabla \varphi$

$\underline{\underline{F}}$ is measure of strain but contains rotations
 $\Rightarrow$ not suitable as strain tensor

Build strain tensor in 3 steps:

- 1) Remove translations
- 2) Remove rotations
- 3) Find principal stretches

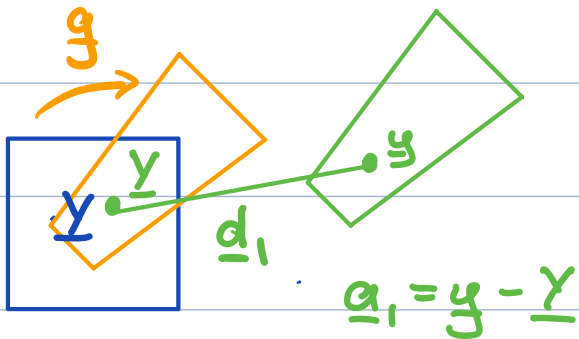
1) Translation - fixed point decomposition

Any hom. φ can be decomposed into

$$\varphi = d_1 \circ g = g \circ d_2$$

$g = \underline{y} + \underline{F}(\underline{x} - \underline{y})$ hom. def. with fixed point $\underline{y}$

$d_i = \underline{x} + \underline{a}_i$ translations from $\underline{y}$ $i = \{1, 2\}$



$$g(\underline{x}) = \underline{y} + \underline{F}(\underline{x} - \underline{y})$$
$$d_1(\underline{x}) = \underline{x} + \underbrace{(\varphi(\underline{y}) - \underline{y})}_{\underline{a}_1}$$

Rewrite φ as Taylor expansion around $\underline{y}$

$$\underline{x} = \varphi(\underline{x}) = \underline{c} + \underline{F} \underline{x} \quad \underline{y} = \varphi(\underline{y}) = \underline{c} + \underline{F} \underline{y}$$

$$\underline{x} - \underline{y} = \underline{F}(\underline{x} - \underline{y})$$

$$\underline{x} = \underline{y} + \underline{F}(\underline{x} - \underline{y}) \quad \text{where } \underline{x} = \varphi(\underline{x}) \text{ and } \underline{y} = \varphi(\underline{y})$$

$$\varphi(\underline{x}) = \varphi(\underline{y}) + \underline{F}(\underline{x} - \underline{y}) \Leftrightarrow \varphi(\underline{x}) = \underline{c} + \underline{F} \underline{x}$$

use to show decomposition $\varphi = d_1 \circ g$

$$\underline{d}_1 = \underline{x} + \underline{\varphi}(\underline{y}) - \underline{y} \quad \text{and} \quad \underline{g} = \underline{y} + \underline{F}(\underline{x} - \underline{y})$$

substitute:

$$\begin{aligned} \underline{\varphi}(\underline{x}) &= \underline{d}_1(\underline{g}(\underline{x})) = \underline{g}(\underline{x}) + \underline{\varphi}(\underline{y}) - \underline{y} \\ &= \cancel{\underline{x}} + \underline{F}(\underline{x} - \underline{y}) + \underline{\varphi}(\underline{y}) - \cancel{\underline{y}} \end{aligned}$$

$$\underline{\varphi}(\underline{x}) = \underline{\varphi}(\underline{y}) + \underline{F}(\underline{x} - \underline{y}) \quad \checkmark$$

$\Rightarrow$ proves $\underline{\varphi} = \underline{d}_1 \circ \underline{g}$

$\Rightarrow$ we can always extract translation
and assume def. with fixed point!

Stretch - Rotation decomposition

$$\text{For } \underline{\varphi}(\underline{x}) = \underline{y} + \underline{F}(\underline{x} - \underline{y}) \quad \text{with} \quad \underline{F} = \underline{R}\underline{U} = \underline{V}\underline{R}$$

$$\underline{\varphi} = \underline{\Gamma} \circ \underline{S}_1 = \underline{S}_2 \circ \underline{\Gamma}$$

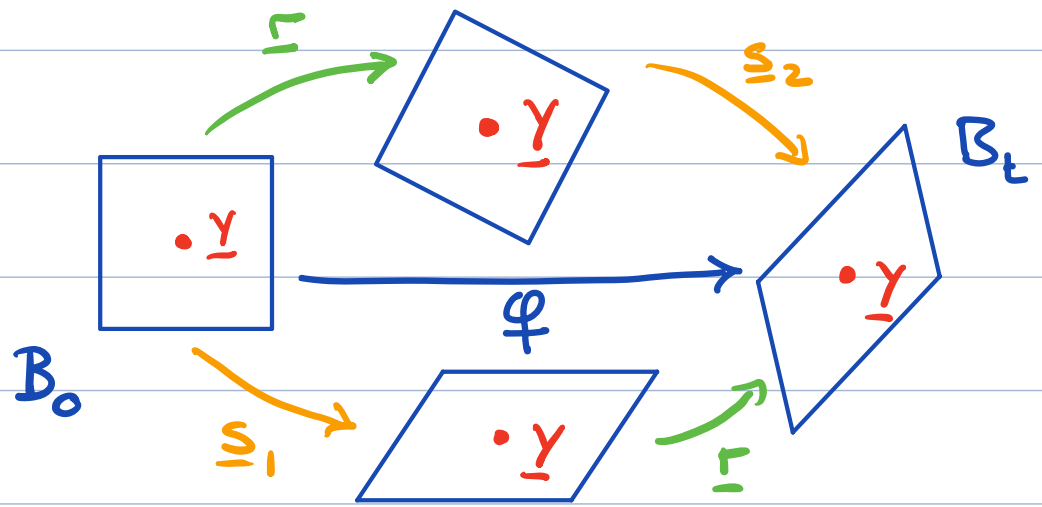
where

$$\underline{\Gamma}(\underline{x}) = \underline{y} + \underline{R}(\underline{x} - \underline{y}) \quad \text{rotation around } \underline{y}$$

$$\left. \begin{aligned} \underline{S}_1(\underline{x}) &= \underline{y} + \underline{U}(\underline{x} - \underline{y}) \\ \underline{S}_2(\underline{x}) &= \underline{y} + \underline{V}(\underline{x} - \underline{y}) \end{aligned} \right\} \text{ stretches from } \underline{y}$$

$$\text{where } \underline{U} = \sqrt{\underline{F}^T \underline{F}} \quad \text{and} \quad \underline{V} = \sqrt{\underline{F} \underline{F}^T}$$

Graphical Interpretation



This can be shown as follows

$$\begin{aligned}
 (r \circ s_1)(\underline{x}) &= r(s_1(\underline{x})) = \underline{y} + \underline{R}(s_1(\underline{x}) - \underline{y}) \\
 &= \underline{y} + \underline{R}(\cancel{\underline{y}} + \underline{u}(\underline{x} - \underline{y}) - \cancel{\underline{y}}) \\
 &= \underline{y} + \underline{R}\underline{u}(\underline{x} - \underline{y}) = \\
 &= \underline{y} + \underline{F}(\underline{x} - \underline{y})
 \end{aligned}$$

$$(r \circ s_1)(\underline{x}) = \varphi(\underline{x}) \quad \checkmark$$

The proof for $\varphi = s_2 \circ r$ is similar.

Stretch tensors

⇒ part of $\underline{\underline{F}}$ that contains deformation

both $\underline{\underline{U}} = \sqrt{\underline{\underline{F}}^T \underline{\underline{F}}}$ and $\underline{\underline{V}} = \sqrt{\underline{\underline{F}} \underline{\underline{F}}^T}$ are s.p.d.

⇒ spectral decomposition

$$\underline{\underline{U}} = \sum_{i=1}^3 \lambda_i \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i \quad \text{and} \quad \underline{\underline{V}} = \sum_{i=1}^3 \lambda_i \underline{\underline{v}}_i \otimes \underline{\underline{v}}_i$$

$(\lambda_i, \underline{\underline{u}}_i)$ eigen pair of $\underline{\underline{U}}$

$(\lambda_i, \underline{\underline{v}}_i)$ eigen pair of $\underline{\underline{V}}$

⇒ same eigen values different eigenvectors

1) Show eigenvalues are same

$$\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}} = \underline{\underline{V}} \underline{\underline{R}} \Rightarrow$$

$$\underline{\underline{R}}^T \underline{\underline{R}} \underline{\underline{U}} = \underline{\underline{R}}^T \underline{\underline{V}} \underline{\underline{R}} \Rightarrow \underline{\underline{U}} = \underline{\underline{R}}^T \underline{\underline{V}} \underline{\underline{R}} \quad \text{change in basis}$$

char. polynomial:

$$\begin{aligned} p_{\underline{\underline{U}}}(\lambda) &= \det(\underline{\underline{U}} - \lambda \underline{\underline{I}}) = \det(\underline{\underline{R}}^T \underline{\underline{V}} \underline{\underline{R}} - \lambda \underline{\underline{R}}^T \underline{\underline{R}}) \\ &= \det(\underline{\underline{R}}^T (\underline{\underline{V}} - \lambda \underline{\underline{I}}) \underline{\underline{R}}) \\ &= \cancel{\det(\underline{\underline{R}}^T)} \det(\underline{\underline{V}} - \lambda \underline{\underline{I}}) \cancel{\det(\underline{\underline{R}})} \\ &= \det(\underline{\underline{V}} - \lambda \underline{\underline{I}}) = p_{\underline{\underline{V}}}(\lambda) \quad \checkmark \end{aligned}$$

$\Rightarrow \underline{\underline{U}}$ and $\underline{\underline{V}}$ same eigenvalues

λ_p 's are principal stretches

$\underline{u}_p$ and $\underline{v}_p$ are right and left principal directions

What is relation between $\underline{u}$ and $\underline{v}$

$$\underline{\underline{U}} \underline{u}_p = \lambda_p \underline{u}_p \quad \text{no sum!}$$

$$\underline{\underline{R}} \underline{\underline{U}} \underline{u}_p = \lambda_p \underline{\underline{R}} \underline{u}_p \quad \underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}} = \underline{\underline{V}} \underline{\underline{R}}$$

$$\underline{\underline{V}} \underline{\underline{R}} \underline{u}_p = \lambda_p \underline{\underline{R}} \underline{u}_p \quad \text{compare} \quad \underline{\underline{V}} \underline{v}_p = \lambda_p \underline{v}_p \quad \text{no sum}$$

$$\Rightarrow \boxed{\underline{v}_p = \underline{\underline{R}} \underline{u}_p} \quad \text{differ by rotation}$$

Summary

Any hom. def. φ can be decomposed into sequence of three elementary deformations:

1) Translation: $\underline{d}_i$ (no strain)

2) Rotation: $\underline{r}$ (no strain)

3) Stretch along principal directions: $\underline{s}_i$ ($\Rightarrow$ strain)

$$\boxed{\varphi = \underline{r} \circ \underline{s}_1 \circ \underline{d}_1 = \underline{d}_2 \circ \underline{s}_2 \circ \underline{r}}$$

Cauchy - Green Strain Tensor

Consider a deformation $\varphi: \mathcal{B} \rightarrow \mathcal{B}'$ with $\underline{\underline{F}} = \nabla \varphi$, then the (right) Cauchy - Green strain tensor is

$$\underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}}.$$

Note that $\underline{\underline{C}}$ is always symmetric pos. definite.

The deformation gradient $\underline{\underline{F}}$ contains information about both rotations and stretches. Using the right polar decomposition we have

$$\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}}$$

$\underline{\underline{R}}$ is rotation matrix

$$\underline{\underline{U}} = \sqrt{\underline{\underline{F}}^T \underline{\underline{F}}} \text{ is right stretch tensor}$$

Clearly $\underline{\underline{C}} = \underline{\underline{U}}^2$ and the rotation $\underline{\underline{R}}$ implicit in $\underline{\underline{F}}$ is not present in $\underline{\underline{C}}$.

$\Rightarrow$ The right Cauchy Green strain tensor only contains information about stretches.

Hence we cannot obtain $\underline{\underline{F}}$ from $\underline{\underline{C}}$!

Remarks:

1) Strictly the right-stretch tensor $\underline{\underline{U}}$ is sufficient.

We introduce $\underline{\underline{C}} = \underline{\underline{U}}^2$ to avoid the tensor square root.

Simple example:

$$[\underline{\underline{F}}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$[\underline{\underline{C}}] = [\underline{\underline{F}}^T][\underline{\underline{F}}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 4 \\ 0 & 4 & 5 \end{pmatrix}$$

To get $[\underline{\underline{U}}]$ we need to solve eigenvalue problem

$$\begin{vmatrix} 1-\mu & 0 & 0 \\ 0 & 5-\mu & 4 \\ 0 & 4 & 5-\mu \end{vmatrix} = (1-\mu)(5-\mu)^2 - 16(1-\mu) = 0$$

Eigenvalues: $\mu_{1,2} = 1$ $\mu_3 = 9$

Eigen vectors: $[\underline{u}_1] = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $[\underline{u}_2] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ $\underline{u}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$$\text{Hence: } [\underline{\underline{U}}] = \sqrt{[\underline{\underline{C}}]} = \sum_{i=1}^3 \sqrt{\mu_i} \underline{u}_i \otimes \underline{u}_i = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$2) \underline{\underline{U}} = \sum_{i=1}^3 \lambda_i \underline{u}_i \otimes \underline{u}_i \text{ where}$$

λ_i 's are principal stretches

$\underline{u}_i$'s are right principal directions

$$\underline{\underline{C}} = \underline{\underline{U}}^2 = \sum_{i=1}^3 \lambda_i^2 \underline{u}_i \otimes \underline{u}_i$$

$\mu_i = \lambda_i^2$ eig. values of $\underline{\underline{C}}$ are squares of principal stretches

eigenvectors are right principal dir.

$$3) C_{KL} = F_{iK} F_{iL} \text{ "material strain tensor"}$$

spatial indices are contracted

Other strain tensors

$$\text{I) } \underline{\underline{E}} = \frac{1}{2}(\underline{\underline{C}} - \underline{\underline{I}}): \text{ Green-Lagrange tensor}$$

$$E_{KL} = \frac{1}{2}(C_{KL} - \delta_{KL}) \text{ material tensor} \Rightarrow \text{linear theory}$$

$$\text{II) } \underline{\underline{b}} = \underline{\underline{F}} \underline{\underline{F}}^T = \underline{\underline{V}}^2: \text{ left Cauchy-Green tensor}$$

$$b_{kl} = F_{kI} F_{lI} \text{ "spatial tensor" (aka. Finger tensor)}$$

$$\text{III) } \underline{\underline{e}} = \frac{1}{2}(\underline{\underline{I}} - \underline{\underline{F}}^{-T} \underline{\underline{F}}^{-1}): \text{ Euler-Almansi tensor}$$

$$e_{kl} = \frac{1}{2}(\delta_{kl} - F_{Ik}^{-1} F_{Il}^{-1}) \text{ "spatial tensor"}$$